

Tuning plant signaling and growth to survive salt

Magdalena M. Julkowska and Christa Testerink

Section of Plant Physiology, Swammerdam Institute for Life Sciences, University of Amsterdam, Amsterdam, Science Park 904, 1098XH Amsterdam, The Netherlands

Salinity is one of the major abiotic factors threatening food security worldwide. Recently, our understanding of early processes underlying salinity tolerance has expanded. In this review, early signaling events, such as phospholipid signaling, calcium ion (Ca^{2+}) responses, and reactive oxygen species (ROS) production, together with salt stress-induced abscisic acid (ABA) accumulation, are brought into the context of long-term salt stress-specific responses and alteration of plant growth. Salt-induced quiescent and recovery growth phases rely on modification of cell cycle activity, cell expansion, and cell wall extensibility. The period of initial growth arrest varies among different organs, leading to altered plant morphology. Studying stress-induced changes in growth dynamics can be used for screening to discover novel genes contributing to salt stress tolerance in model species and crops.

How flexibility enables increased abiotic stress tolerance

Plants are flexible organisms, with the potential to develop a plethora of morphological patterns depending on the growth conditions to which they are exposed. This morphological flexibility has enabled plants to colonize almost every corner of the globe and to survive in the harshest conditions. Today, the demand for food is on the rise, while arable lands are becoming scarce [1]. To meet future demands, an increase in plant yield in environmentally challenging conditions is essential.

High soil salinity is one of the major abiotic factors limiting crop yield. Halophytes have evolved functional and structural adaptations, such as excretion of salt through secretory glands, ensuring their survival in saline environments [2]. Most crop plants are glycophytes, which are not able to withstand high levels of salt stress and do not have morphological structures allowing salt excretion. By contrast, glycophytes exhibit natural variation in their salinity tolerance between as well as within plant species. This natural variation can be used for the identification of genetic components underlying salinity tolerance in glycophytes [3,4]. Populations comprising natural accessions of

model and crop species, such as *Arabidopsis* (*Arabidopsis thaliana*), rice (*Oryza sativa*), and maize (*Zea mays*), are now broadly used for genome-wide association studies (GWAS). Intraspecific variation can also be a valuable source of allelic variation in previously characterized molecular players, leading to new insights into the salinity tolerance mechanisms [5]. To produce new crop varieties, with reliable and high yield in saline environments, the physiology and allelic variations underlying local adaptations to salinity stress need to be understood.

Salt stress limits plant growth by increasing the osmotic potential of the soil and, thus, decreasing water uptake by the roots. In the long term, accumulation of sodium ions (Na^+) in the shoot compromises plant growth by reducing photosynthesis rate if Na^+ are not compartmentalized at the cellular or intercellular level. Growth of individual plant organs is reduced by salinity stress with varying magnitude, leading to altered general plant morphology, such as a change in the root:shoot ratio [3]. These salt-induced changes in plant morphology are likely to affect the performance of a plant under saline conditions. Recent discoveries in the field of salt stress physiology provide increased understanding of plant acclimation responses resulting in altered plant development and ultimately increased salinity stress tolerance. In this review, we summarize the signaling pathways crucial for early salt stress signaling and the acclimation responses leading to altered plant morphology, which significantly affects plant performance under salt stress conditions.

First aid: early signaling responses to salt stress

The first reactions of a plant to salinity occur within seconds to hours upon exposure to salt stress. An overview of early signaling responses is presented in Figure 1. Sodium ions enter the root epidermal and cortical cells through nonselective cation channels (NSCCs) [6]. The influx of Na^+ induces depolarization of the plasma membrane, activating potassium (K^+) outward-rectifying channels (KOR) and reducing net passive potassium uptake through inward-rectifying K^+ channels [7].

Salt stress additionally causes an increase in osmotic soil potential, resulting in water loss and reduction in turgor pressure. One of the mechanisms potentially involved in sensing changes in turgor pressure involves activation of mechanosensitive receptor kinase cyclase, resulting in rapid (<5 s) accumulation of cGMP, which inhibits sodium influx by deactivation of NSCC channels [8]. Accumulation of cGMP is also thought to underlie

Corresponding author: Testerink, C. (C.S.Testerink@uva.nl).

Keywords: salt stress; developmental plasticity; cellular signaling; Root System Architecture; osmotic stress; potassium starvation; root development; natural variation.

1360-1385/

© 2015 Elsevier Ltd. All rights reserved. <http://dx.doi.org/10.1016/j.tplants.2015.06.008>

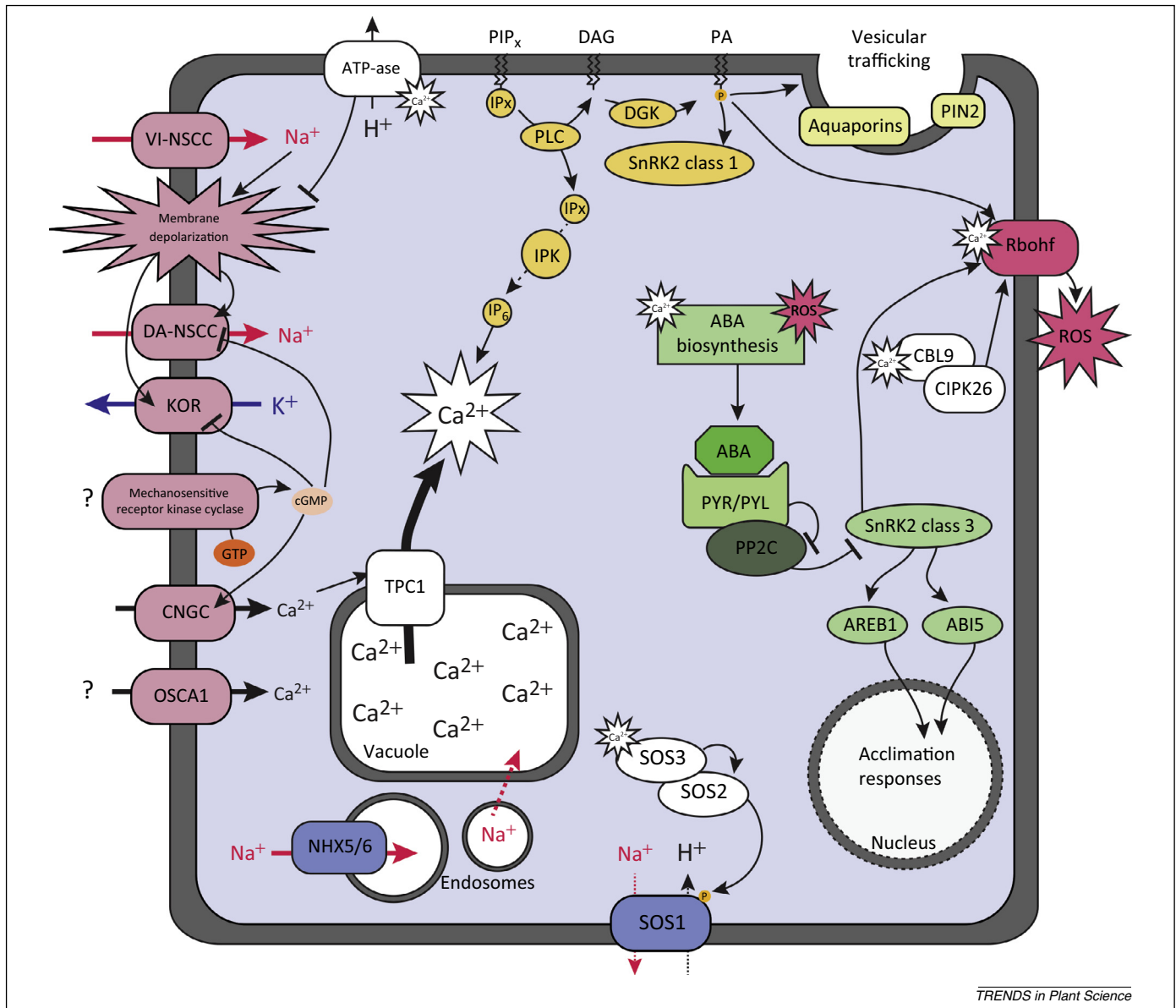


Figure 1. Early signaling in response to salinity stress. Upon salt stress exposure, sodium ions (Na⁺) enter the cell through voltage independent nonselective cation channels VI-NSCCs, inducing membrane depolarization [6], depolarization activated NSCC (DA-NSCC) and potassium ion (K⁺) outward rectifier channels (KOR) [7]. Additionally, changes in turgor pressure might be sensed through mechanosensitive receptor kinase cyclases, inducing production of cGMP or reduced hyperosmolality-induced calcium ion ([Ca²⁺]) increase 1 (OSCA1), a plasma membrane-localized Ca²⁺ channel [10]. cGMP inhibits KOR and induces opening of cyclic nucleotide gated channels (CNGC), responsible for apoplasmic Ca²⁺ influx [9]. Increased cytosolic Ca²⁺ release is accomplished by activation of slow-activating vacuolar/two pore channel 1 (SV/TPC1) [12]. High cytosolic Ca²⁺ activates among others calcium-binding calmodulin (CBL9). This interacts with calcium-induced protein kinase (CIPK26), which targets respiratory burst oxidase homolog F (RbohF), inducing production of reactive oxygen species (ROS) [86]. Salt stress-induced phospholipase C (PLC) contributes to production of phosphatidic acid (PA) via phospholipase D (PLD)/diacylglycerol kinase (DGK) pathways, as well as an increase in inositol phosphate 6 (IP₆) through phosphorylation of inositol phosphates (IP) by IP kinase [29]. The increase in PA affects vesicular trafficking of PIN-FORMED 2 (PIN2) [76] and possibly aquaporins, abscisic acid (ABA)-independent sucrose non-fermenting like kinase 2.6 (SnRK 2.6) class 1 proteins, and the activity of RbohF [31]. An increase in cytosolic Ca²⁺ and ROS induces ABA accumulation [21], which is sensed by pyrabactin resistance (PYR)/PYR1-like (PYL) receptors, releasing ABA-dependent protein phosphatase 2C (PP2C) from ABA-dependent SnRK2 class 3, which in turn activates the transcription factors ABA responsive element-binding protein (AREB1) and ABA-insensitive 5 (ABI5) and increases ROS through activation of RbohF [25]. Reduction of cytosolic sodium is guided through activation of the salt overly sensitive (sos) signaling pathway [34], compartmentalization of sodium in endosomes by Na⁺/H⁺ exchanger 5/6 (NHX5/6) [37], and possibly subsequent fusion with the vacuole. Positive and negative regulatory actions are indicated by arrows and lines with bars, respectively.

initial Ca²⁺ signal mobilization through activation of cyclic nucleotide gated channels (CNGC) [9] allowing influx of apoplasmic Ca²⁺ into the cytosol. Another osmo-sensing mechanism involves reduced hyperosmolality-induced [Ca²⁺] increase 1 (OSCA1), a plasma membrane-localized protein that is responsible for increased Ca²⁺ influx in response to osmotic stress [10]. However, both proposed osmo-sensing mechanisms are yet to be validated for their

role in salt stress-induced osmotic signaling and induction of Ca²⁺ influx.

The initial increase in cytosolic Ca²⁺ occurs within seconds after application of salt stress and is further facilitated through activation of slow-activating vacuolar/two pore channel 1 (SV/TPC1), resulting in release of vacuolar Ca²⁺ [11]. In response to local salt treatment, mobilization of vacuolar Ca²⁺ through TPC1 results in

rapid (400 $\mu\text{m/s}$) Ca^{2+} signal propagation to other parts of the root and shoot [12]. The cytosolic Ca^{2+} increase is predominantly enriched in cortex and endodermis layers of the roots. High cytosolic Ca^{2+} activates Ca^{2+} /calmodulin (CaM)-dependent kinases, which in turn stimulate among others plasma-membrane H^+ -ATPases, restoring membrane voltage and inhibiting depolarization-activated NSCCs (DA-NSCCs) [13,14,86]. Production of ROS is also stimulated by increased cytosolic Ca^{2+} [15,16]. Propagation of the ROS signal depends on activation of respiratory burst oxidase homolog F (RbohF) [17], which is controlled by ethylene [18], ABA-dependent protein kinase sucrose non-fermenting like kinase 2.6 (SnRK 2.6) [19] and phosphatidic acid (PA) binding [20]. Both ROS and Ca^{2+} modulate the release of ABA (reviewed in [21]), leading to initiation of the transcriptional responses.

A rapid increase in ABA is reached through the cleavage of carotenoid precursors by dioxygenases, with 9-*cis*-epoxycarotenoid dioxygenase 3 (NCED3) having an essential role in salt stress-induced ABA accumulation [22]. In roots, ABA production is additionally enhanced by an increase in the synthesis of the carotenoid precursors [23]. The increase in ABA occurring within hours after application of salt stress [24] is sensed by pyrabactin resistance (PYR)/PYR1-like (PYL) receptor proteins, which bind upon activation to protein phosphatase 2C (PP2C), releasing the inhibition of downstream protein kinases, such as class 3 SnRK2. The ABA-dependent SnRK2s, OST1/SnRK2.6, SnRK2.3, and SnRK2.2, target transcription factors [ABA responsive element-binding protein (AREB1) and ABA-insensitive 5 (ABI5)], which activate genes involved in not only acclimation responses, but also plasma membrane ion channels [slow anion channel associated 1 (SLAC1) and potassium channel (KAT1)] and the NADPH oxidase RbohF [25]. Activation of ABA-dependent signaling underlies stomatal closure and has an important role in regulating growth and acclimation responses, which are discussed below.

The other members of the SnRK2 family are also activated upon salinity stress; however, their activation is not solely dependent on ABA [26]. Subclass 1 SnRK2s (SnRK2.1, SnRK2.4, SnRK2.5, SnRK2.9, and SnRK2.10) are induced upon hyperosmotic stress in an ABA-independent manner [26]. Interestingly, two subclass 1 SnRK2s (SnRK2.4 and SnRK2.10) were observed to bind directly to the signaling phospholipid, PA [27,28]. Production of PA is rapidly induced upon exposure to salinity through the hydrolysis of phosphatidylinositol-phosphate phospholipids (PPIs) by phospholipase C (S.A. Arisz, PhD thesis, University of Amsterdam, 2010) into inositol phosphates (IP_x) and diacylglycerol (DAG). Both IP_x and DAG undergo further phosphorylation by IP - or DAG kinases into IP_6 and PA, respectively [29]. While IP_6 causes an increase in cytosolic Ca^{2+} [30], the increase in PA alters membrane structure, as well as being an important signaling molecule by itself [31]. PA facilitates protein recruitment to cellular membranes, allowing interactions between its molecular partners. PA-binding proteins include clathrin assembly proteins (AP180/ECAs), RbohF, ABI1 and a potassium channel β subunit [20,31,32], suggesting the importance of membrane-based PA in

salt stress-induced vesicular trafficking, ROS burst and K^+ homeostasis.

After an initial increase in Na^+ in the plant root, a decrease in cytosolic Na^+ is likely mediated through the action of the salt overly sensitive (SOS) pathway. Salt stress reduces the interaction between the protein kinase SOS2 and λ or κ 14-3-3 proteins [33], allowing SOS2 interaction with calcium-induced calcineurin B-like family protein SOS3/CBL4 [34]. The SOS3/SOS2 complex activates the putative Na^+/H^+ exchanger 7 (NHX7/SOS1), which removes Na^+ from the cytosol into the apoplast of root cortex and epidermis, although the exact location of SOS1 in the root remains to be established [35,36]. Additionally, sodium compartmentalization into the vacuole by NHX transporters reduces cytosolic Na^+ concentrations and maintains turgor pressure of the vacuole negative osmotic potential [37]. The exact mechanisms of sodium loading into the vacuole remain ambiguous. Mutants in vacuolar NHX1/2 transporters are severely impaired in vacuolar K^+ acquisition and are rescued by salt stress treatment, supposedly restoring vacuolar potential for cell expansion [38]; yet, there is no direct evidence for sodium vacuolar transport by NHX1/2. By contrast, endosomal NHX5/6 mutants exhibit severe sensitivity to salt stress [37], suggesting that Na^+ loading into the vacuole is regulated through endosomal Na^+ scavenging and subsequent fusion to the vacuole [39].

Stop, adjust, and march on: salt stress-induced growth regulation

Exposure to salinity stress results in the initiation of a quiescent state, which is followed by growth recovery, albeit often at lower growth rates than observed under control conditions (summarized in Figure 2). The initiation of quiescence is thought to be independent of sodium accumulation in photosynthetic tissues and, therefore, is ascribed to the 'osmotic' effects of salt stress [40]. Interestingly, drought stress was also observed to induce growth quiescence, which is mediated by increased levels of ethylene, activating ethylene responsive transcription factors ERF5 and ERF6, which in turn induce transcription of the gibberellin-degrading enzyme gibberellin 2-oxidase 6 (GA2ox6) [41]. Although an ERF5/ERF6 double mutant exhibited lower growth inhibition compared with wild-type plants in drought conditions, no differences were observed for salt stressed plants [41]. Those results illustrate that, although initial growth reduction in salt stress conditions is independent of a build up of sodium in shoot tissues, it is regulated by mechanisms that might be distinct from osmotic stress. Exposure to salt stress induces an increase in ABA levels, which results in ABI1-dependent stabilization of DELLA proteins [42], whereas repression of brassinosteroid signaling further enhances growth quiescence [24]. Stress-induced repression of the gibberellin and brassinosteroid signaling pathways results in lower cell cycle activity and could underlie a reduction in meristem size during the quiescent phase [43]. Additionally, in the elongation zone, turgor pressure within the cell decreases as salt stress increases the osmotic potential of the soil, leading to loss of water from cells. The maintenance of turgor pressure can be facilitated by compartmentalization

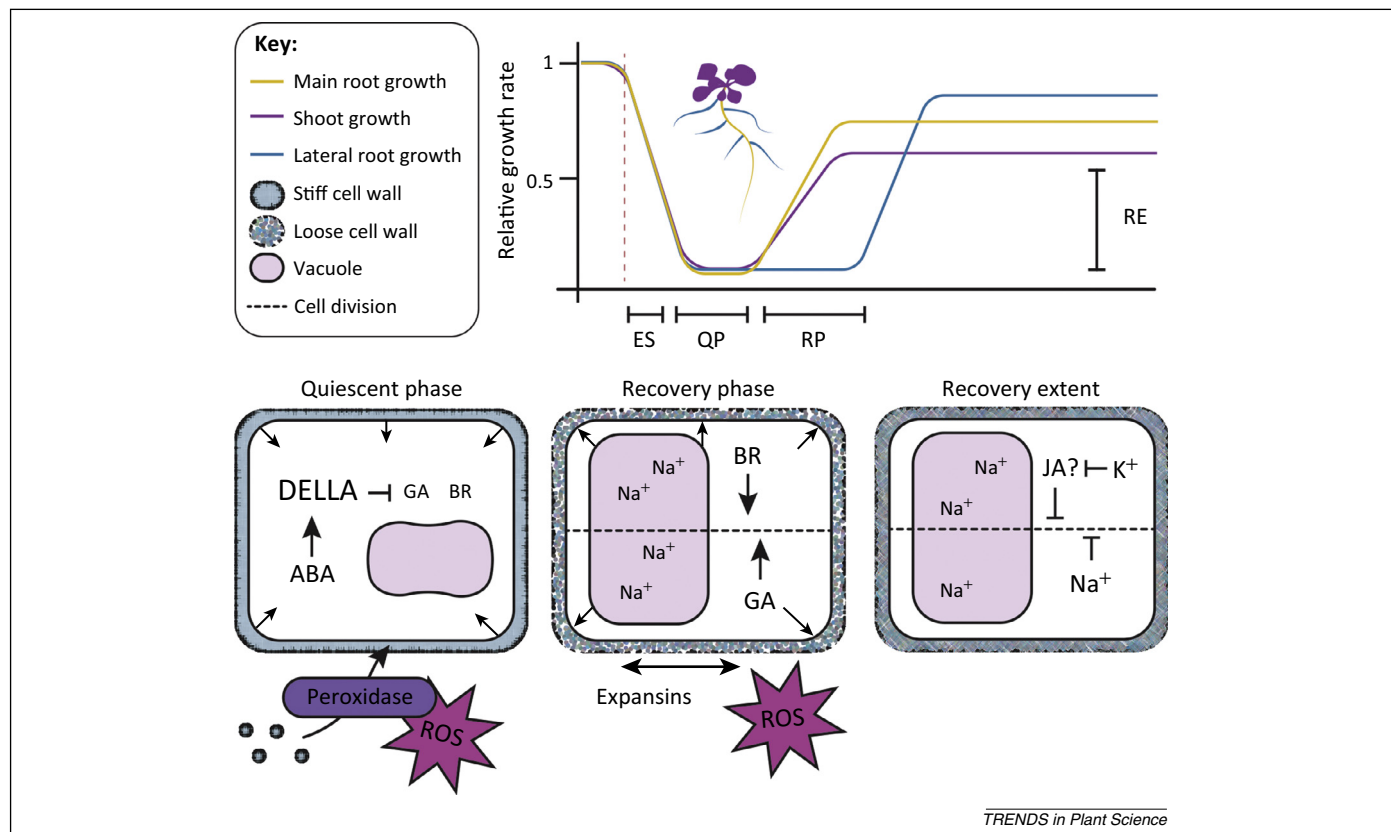


Figure 2. Growth regulation during exposure to salinity stress. Exposure to salt stress (indicated by the vertical red-broken line) results in activation of early signaling (ES), resulting in a rapid reduction in growth rate and induction of the quiescent phase (QP). Subsequently, transcriptional regulation induces a growth recovery phase (RP). The duration of the QP and recovery extent (RE) depends on the plant organ, local sodium concentrations, and sensitivity to salt stress [24,64,60]. The QP of lateral roots (blue line) is longer than the QP of the main root (yellow line) [24,64]. Given that the root:shoot ratio increases after exposure to salt stress, RE of the shoot growth (purple line) is likely to be lower than RE of the root. In the QP, growth is restrained by reduction of cell division through abscisic acid (ABA)-dependent stabilization of DELLA proteins [85], which inhibit gibberellin (GA) signaling, as well as downregulation of brassinosteroid signaling (BR) [24]. Cell expansion is reduced by decreased turgor pressure as well as increased cell wall stiffness induced by high apoplastic reactive oxygen species (ROS) acting in concert with peroxidases in cross-linking phenolic compounds and glycoproteins of the cell wall [49]. In the subsequent RP, sodium ions (Na⁺) are compartmentalized into the vacuole, enhancing turgor pressure. Cell cycle activity is recovered by removing initial restraints on BR and GA signaling [51]. Cell elongation is aided by an increase in cell wall extensibility through cleavage of polysaccharides by ROS and expansins, resulting in cell wall loosening. The RE is restrained by the jasmonic acid (JA) signaling pathway [58], which might be activated by high Na⁺:potassium ion (K⁺) ratios resulting in responses similar to potassium starvation [57]. Both high sodium concentration and JA result in inhibition of cell cycle machinery [43,58].

of Na⁺ into the vacuoles, which increases the osmotic potential of the vacuole in a cost-effective manner [37].

Downregulation of aquaporin expression upon salt treatment limits turgor maintenance and, therefore, cell expansion [44]. The decrease in water conductance together with stomatal closure reduces the transpiration rate and, consequently, may reduce sodium accumulation in shoot tissues [45]. In the short term, hydraulic conductivity of the roots is regulated by internalization of aquaporin proteins [46]. Long-term regulation of hydraulic conductivity is ascribed to morphological adaptations, such as increased thickness of the Casparian strip [47] or a reduction in stomatal development [48]. Additionally, salt stress-induced cell wall stiffening limits cell elongation [49]. The key players in this process are peroxidases present in the apoplast, which use ROS as a co-substrate. During the early stages of stress, high apoplastic ROS levels induce peroxidase activity. Peroxidase cross-links phenolic compounds and glycoproteins of the cell wall, resulting in reduced extensibility of the cell wall and limited cell expansion [49].

After the quiescent period, plants exposed to salinity stress enter the growth recovery phase. Formation of stress granules (SGs), a group of cytoplasmic foci that assemble in

response to stress, has an important role in this process. SGs act as centers of messenger (m)RNA regulation through selective re-initiation or degradation of mRNA [50]. Salt stress-induced SGs stabilize the transcripts of GA20ox3, a key enzyme for gibberellin biosynthesis [51], allowing reactivation of cell cycle activity. Suppression of gibberellin biosynthesis with paclobutrazol prevented growth recovery of *Arabidopsis* roots grown under saline conditions [24]. Surprisingly, inhibition of ABA synthesis by fluridone treatment also ceased the growth recovery [24], suggesting a crucial role of ABA in growth recovery upon salinity stress. With prolonged exposure to salt stress, the extensibility of the cell wall increases. During the later stages of salt stress, where the availability of phenolic compounds and glycoprotein is low, high ROS levels in the apoplast result in increased OH⁻ radicals, which cleave sugar bonds in polysaccharides in the cell wall. This together with activity of xyloglucan-modifying enzymes and expansins present in the apoplast result in cell wall loosening, allowing growth recovery [49].

The extent to which growth rates recover during the later stages of salinity stress relies on hormonal cues, sensitivity of early signaling, and the capacity of the plant for acclimation [3]. Exclusion of sodium from leaf tissue

contributes to salt stress tolerance. The product of the gene encoding high affinity K^+ transporter 1 (*HKT1*) is a plasma membrane importer that transports Na^+ and, therefore, has an important role in sodium exclusion. Root stele-specific overexpression of *HKT1* in *Arabidopsis* plants resulted in an increased efficiency of Na^+ retrieval from the xylem, thus reducing Na^+ accumulation in leaf tissue and yielding larger rosettes in the presence of salt [52].

An excess of Na^+ negatively impacts the acquisition and balance of essential nutrients, such as K^+ , Ca^{2+} , and magnesium (Mg^{2+}) [53]. Due to the physiochemical similarities between Na^+ and K^+ (reviewed in [54]), Na^+ tend to substitute K^+ at its usual binding sites and impair cellular biochemistry. Increasing K^+ concentrations can alleviate the cytotoxic effects of Na^+ during salt stress, which is exemplified by partial rescue of *sos* mutant phenotypes [55] by application of an additional K^+ source. Increased potassium acquisition in polyploid *Arabidopsis* plants corresponded to enhanced salt stress tolerance [56]. By contrast, potassium starvation induces transcriptional changes related to jasmonic acid (JA) signaling [57]. Interestingly, activation of JA signaling was also observed to modulate the transcriptional programs during the later stages of salt stress exposure [24]. Activation of JA signaling results in decreased meristem size and cell elongation [58] and could determine the recovery of growth rate during the later stages of salt stress, because the JA-insensitive mutant *jai3-1* showed higher rates of recovered growth [24]. The mechanisms guiding the maintenance of the $Na^+ : K^+$ ratio during the later phases of salinity stress remain unclear. Interestingly, K^+ starvation reduces root slanting [59], a response that is also observed upon exposure to salt stress [60]. In addition, the *akt1* mutant, which is deficient in high-affinity K^+ uptake, showed reduced root slanting under control conditions, suggesting that root slanting is a reliable read-out of potassium acquisition [59]. Quantitative trait locus (QTL) analysis of RSA responses to K^+ starvation revealed one locus on chromosome 1 corresponding to main root angle [61]; however, the underlying genes have yet to be identified through further fine mapping.

Although many of the signaling pathways leading to alteration of growth are shared among different plant organs, the impact on the growth rate of individual organs and developmental processes differs. Germination of *Arabidopsis* seeds is inhibited by concentrations exceeding 150 mM NaCl, while shoot and root growth are reduced by half at 50 and 100 mM NaCl, respectively [62]. An increase in the root:shoot ratio has long been hypothesized to contribute to salinity tolerance, because it would limit water loss through stomata and enhance water uptake by extending the size of the root system [3]. To date, the molecular players underlying differential responses of organs to stress are still to be unraveled, but are likely to depend on tissue context, stress gradients, and cell type-specific signals perceived and generated at individual levels of functional plant organization [63]. Comparing the growth rates of roots and shoots and identification of the genetic basis for the phenomenon requires sophisticated phenotyping platforms. Recently, different parts of the root were found to respond with varying magnitudes

of growth inhibition to salinity stress [60,64]. Given that a plant root system exhibits clear hierarchy and spacing of developmentally distinct regions, it seems a perfect model for studying the processes involved in stress-induced growth regulation and the consequences of morphological acclimation responses in the context of salinity tolerance.

Back to the roots: salt stress altering root development Signals controlling root system architecture

The root system architecture (RSA) of dicotyledonous plants comprises an embryonically derived main root (MR) and postembryonic lateral roots (LRs), which can further branch into higher-order LRs. Under control conditions, RSA development is mainly guided by the antagonistic action of two hormones: auxin and cytokinin [65]. A schematic overview of signals controlling root development under control conditions is presented in Figure 3A. Auxin is distributed along the root in a gradient, with the highest auxin concentration found in the root apical meristem, enhancing cell cycle activity and reducing cell elongation. The regular oscillations in auxin levels at the root tip initiate the formation of founder cells in xylem pole pericycle cells, which form the starting point of LR development [66]. Nonprimed pericycle cells are arrested in the G1 mitotic phase, whereas founder cells undergo simultaneous activation of cell cycle and cell fate genes triggered through auxin-dependent degradation of single root/indole-3-acetic acid inducible 14 (SLR/IAA14) [67]. Although cell division and LR morphogenesis both rely on auxin signaling, they are regulated independently. *Diageotropica/wol* mutants of tomato (*Solanum lycopersicum*) and *Arabidopsis*, respectively show maintained proliferative capacity of pericycle cells, whereas the LR development is arrested and cannot be rescued by additional auxin application [68,69]. Additionally, cells surrounding the developing LR primordia (LRP) undergo dramatic morphological changes. Deflating the cells in direct contact with developing LRP and localized breaks in the Casparian strip allow emergence of LRs without significant gaps in the apoplastic barrier [70]. The initial formation of LRP is inhibited by cytokinin, which prevents cell cycle re-entry of the founder cells in direct proximity of already established LRs [71,72], ensuring spatial distribution of LRs and efficient exploration of soil environment. During the later stages of LRP development cytokinin promotes cell differentiation and elongation [65]. The integration of auxin and cytokinin antagonistic mechanisms might be mediated by gibberellins, because auxin induces degradation of DELLA proteins and enhances cell cycle activity, whereas gibberellins limit the growth inhibition mediated through cytokinin (reviewed in [65]).

RSA modulation in response to salt stress

Mild salinity stress was observed to enhance the growth of MR and LR, while higher levels of salt were detrimental for both [73,74]. The main signals controlling root development under salt stress conditions are presented in Figure 3B. Development of RSA in saline stress conditions is modified through the action of ABA, ethylene, JA, auxin, and brassinosteroids [24,42,64,73,75]. Salt stress results in a reduction of MR meristem size because the number of

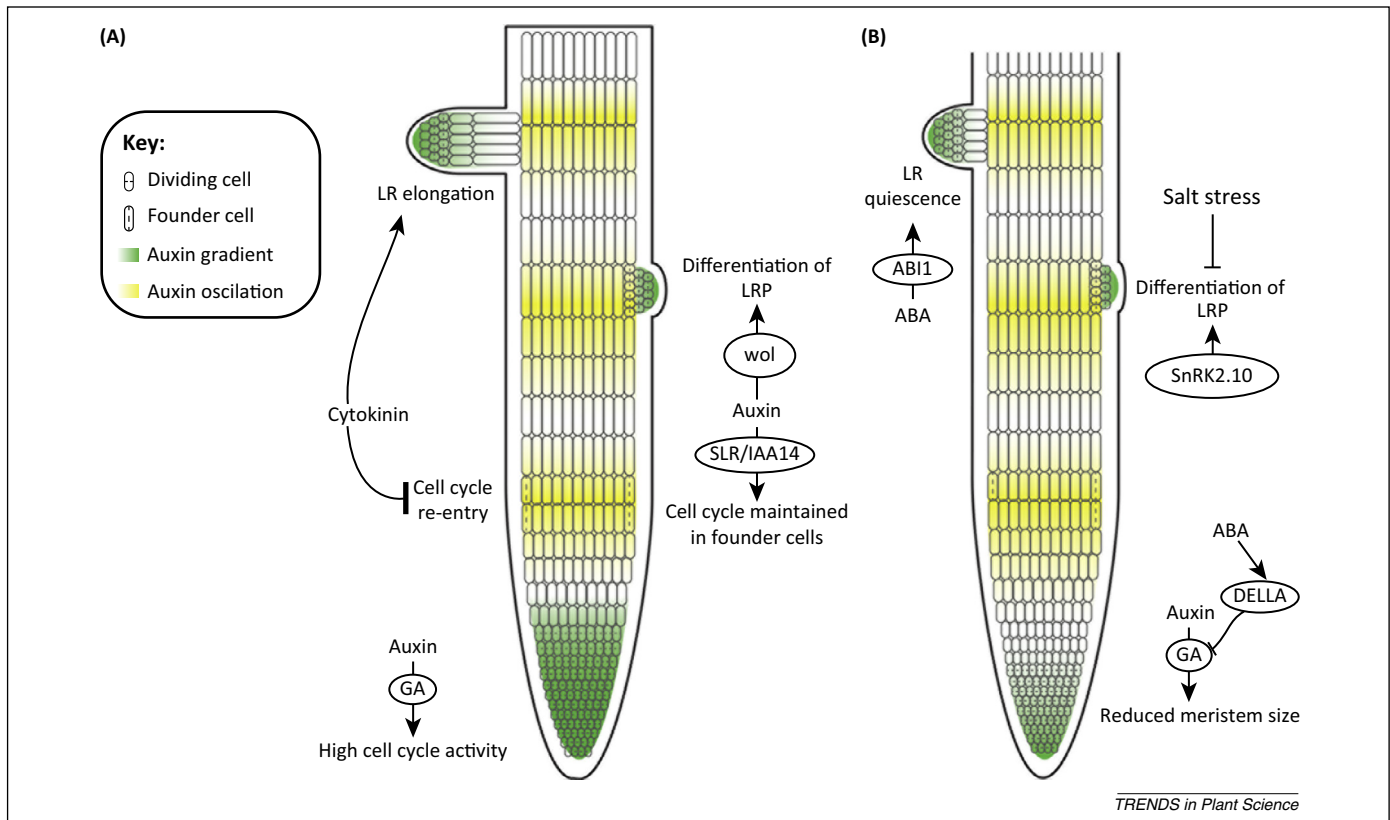


Figure 3. Regulation of root development under control and salt stress conditions. Root development under control conditions (A) depends on an auxin gradient along the root, which induces cell division and represses cell expansion [65]. Auxin oscillations at the root tip result in the formation of founder cells in the xylem pole pericycle, which are the starting point for lateral root (LR) formation [66]. Through induction of cell division, dependent on single root (SLR/IAA14) [67], and cell differentiation, relying on wol [68], LR primordia (LRP) develop into LRs. Cytokinin inhibits cell cycle activity as well as primordium development and stimulates cell elongation in developing LR. Under salt stress conditions (B), the meristem size is limited by stabilization of DELLA proteins, inhibiting cell division [85]. Although LR primordium development is reduced by salt stress, sucrose non-fermenting like kinase 2.10 (SnRK2.10) promotes LR development [28]. The newly emerged LRs undergo a quiescent stage, which is regulated in an abscisic acid (ABA)-dependent manner [64]. Abbreviation: GA, gibberellin.

cells involved in cell division decreases [43]. Salt stress-induced ABA is produced in the early maturation zone [24], leading to inhibition of gibberellin signaling. Development of LRP depends on the salt stress levels. While 50 mM NaCl increased the portion of LRP that developed into LRs [73], higher levels of salt stress reduced development of LRs from LRP [28]. A more severe reduction in LRP development was observed in the *snrk2.10* mutant [28]. Thus far, no studies have addressed patterning of LRP or auxin oscillations leading to altered LR patterning in response to salt stress.

Arabidopsis, tomato, and sorghum (*Sorghum vulgare*) plants exposed to a salt gradient exhibited a negative halotropic response of the MR, prioritizing salt avoidance over gravitropic growth [76]. This response is specific to Na^+ and is mediated through differential auxin distribution, resulting in root bending away from salt, which is accomplished by internalization of the auxin efflux carrier PIN-FORMED 2 (PIN2) at the side of the root facing higher salt concentrations. The redistribution of PIN2 relies on phospholipase D-dependent clathrin-mediated endocytosis, implying an important role of phospholipid signaling during the early events of salinity sensing. Studying the molecular mechanisms behind the halotropic response could lead to identification of the early components in salt stress signaling and might shed more light on early signaling and sodium-sensing machinery.

The activation and propagation of initial Ca^{2+} signals [12] and the majority of the short-term transcriptional responses [24] predominantly occur in the root endodermis, where environmental and developmental signals are integrated (reviewed in [77]). The location of the endodermis layer, surrounded by the Casparian strip, which limits apoplastic ion transport, facilitates stress signal recognition and activation of signaling cascades only when the ion build-up reaches critical concentrations on the inside of the plant. Given that the Casparian strip gets thicker after exposure to salt stress, limiting the influx of Na^+ ions to the inner layers of the root [47], the sensing of salt stress in the endodermis can be adjusted as morphological adaptations occur. Salt stress-induced endodermal ABA signaling was found to underlie the quiescence of newly emerged LRs, which is one of the first RSA responses to salt stress [64]. Quiescence of LRs correlates with downregulation of the cell cycle marker cyclin B1 (CYCB1), implying that lateral root meristem activity is repressed. Mis-expression of *abi1-1*, resulting in dominant suppression of ABA signaling, leads to loss of salt-induced quiescence and to collapse of LR growth during the later stages following salt stress exposure (136 h post transfer). Interestingly, ABA-induced quiescence of LRs depends on a developmental switch in meristem sensitivity, because further-developed LRs exhibited salt and ABA responses similar to the MR [64]. Exposure to salt stress can also alter the

developmental fate of the cells. Root hair outgrowth is suppressed upon salt treatment through reprogramming of root-hair cells by ectopic expression of GLABRA2 and repression of root-hair cell identity [78].

It is complicated: relation between RSA and salt stress tolerance

The architecture of the root is likely to affect the development of the shoot in control and saline conditions [63]. In most *sos* mutants, reduced root growth corresponds to increased salt stress sensitivity [18,79,80]. Studies in rice suggest an important role for LR formation and emergence in sodium exclusion, because Na^+ were observed to leak through LR branching sites, suggesting that plants with a reduced number of LRs would have an advantage in terms of efficient Na^+ exclusion [81,82]. Although the molecular regulation of LR emergence ensures minimal damage in the apoplastic barrier [70], the root-branching sites might be the weakest spots for apoplastic ion exclusion. Additionally, the expression pattern of the sodium transporter HKT1 suggests that efficient ion compartmentalization occurs in the root stele of fully matured roots [83,84]. Taken together, the results suggest that RSA with relatively few but longer LRs might be the most successful strategy in sodium exclusion and compartmentalization in the root stele.

To examine the correlation between RSA and salinity tolerance, the salt-specific alterations in root development were determined by capturing dynamics of individual root architecture components in descriptive models. The use of descriptive models allowed a comparison of the effect of salt stress on individual root components by looking at the relative effect of salt stress, and showed that the growth of MR was inhibited more severely compared with LR development in the *Arabidopsis* Col-0 accession [60]. The effect of salt on MR and LR development is dependent on the time and spatial scale of the observed response. On shorter timescales (<72 h), the prolonged quiescence of LR dominated the salt stress-induced response [64], while on a longer timescale (12 days), the development of LR was recovered while the MR growth was impaired more severely compared with LR development [60]. Extending the study to a diversity collection of natural accessions, further enabled identification of four major strategies in *Arabidopsis* accessions, because the developmental differences observed between individual accessions at control conditions were normalized. Interestingly, accessions that combined a severe decrease in LR elongation with a minor decrease in MR or LR number formation, exhibited the smallest increase in the $\text{Na}^+:\text{K}^+$ ratio upon salt stress exposure [60]. Those results are contradictory to the studies conducted in rice [81,82], where reduced LR emergence was observed to positively correlate with Na^+ exclusion. This difference could be due to anatomical differences between *Arabidopsis* and rice roots, such as aerenchyma and exodermis tissues. The inhibition of LR growth in *Arabidopsis* in the most tolerant accessions could be due to a prolonged quiescent phase of the LRs. The relation between growth quiescence and salinity tolerance was illustrated in earlier studies, where stabilization of DELLA proteins resulted in enhanced salinity tolerance

[85], whereas the release of initial LR quiescence in *abi1-1* mutants resulted in growth collapse during the later stages of salinity stress [64]. Further validation of this relation, as well as identification of genetic components responsible for salt-induced alteration of RSA, will provide more understanding of the RSA contribution to salinity tolerance. The correlation between RSA strategies and ion accumulation implies that the mechanisms involved in LR quiescence, rather than LR emergence, are crucial for RSA-derived salinity tolerance.

Although the quiescence of both MR and LR is regulated by ABI1-dependent DELLA stabilization [64,42], the duration of the quiescent phase is more than ten times as long for newly emerged LR than for the main root (MR) (80 h versus 7 h) [24,64]. This suggests that the differential growth observed in MR and LRs [61] is due to either the duration of the quiescence phase or the extent of growth recovery. The mechanisms responsible for differential growth regulation could be caused by local differences in sodium accumulation between the individual components of RSA. Other mechanisms, different from sodium accumulation, might be also important for differential growth regulation, because the sensitivity of MR and LR meristems to ABA and ethylene differs between not only the meristems, but also individual accessions [60,64]. Growth recovery and differential growth regulation leading to morphological changes depending on hormone level and/or sensitivity, local sodium accumulation, and signaling sensitivity, remain to be established. The concept of differential regulation of RSA dynamics is novel [60] and should be further exploited by using natural variation and available genetic tools to determine molecular players in the observed strategies among the accessions.

Concluding remarks

Salinity stress is widely described to reduce plant growth and to affect plant development. The impact of salt stress on the growth of individual plant organs differs in severity, resulting in altered plant morphology. Identification of phenotypic traits with limited genetic complexity and strong correlation to salinity tolerance is crucial for the successful discovery of novel candidate genes and allelic variation therein. The recent studies summarized here suggest that relative changes in growth morphology are important contributors to salinity tolerance. Capturing growth dynamics in descriptive models [60] focusing on different time and spatial scales could be extrapolated to the shoot tissue for more efficient identification of genes relevant for salt-specific modifications in plant morphology. Although early signaling responses that occur within seconds to hours after salt stress exposure are relatively well characterized, the temporal data on activation of individual signaling pathways is often lacking. By characterizing the timing of individual responses, the growth-related phenotypes at various timescales could be linked to specific processes, such as ABA synthesis or subcellular sodium compartmentalization (Box 1). More precise quantification of growth responses will further establish whether differentiation in organ growth rates is due to either variation in lengths of the quiescent phases or the extent of growth recovery. Although quantification of growth

Box 1. Outstanding questions

- Early responses to salt stress are relatively well defined, but what are the temporal patterns of activation and deactivation of individual signaling pathways and how do those patterns relate to spatiotemporal alterations of plant growth?
- What are the factors that determine the differential sensitivity of individual plant organs to salt? Does sensitivity rely on differential salt accumulation or tissue-specific induction of signaling?
- Are the signaling pathways underlying the osmotic phase of salt stress distinct from hyperosmotic stress? What are the genetic components of osmotic signaling in salt stress responses?
- What are the processes underlying growth recovery following the initial quiescent phase? Does local potassium starvation have a role in growth recovery?
- How are Na⁺ ions perceived by the plant root?
- How does growth modulation in response to salinity contribute to salt tolerance?

dynamics is time consuming, it provides novel insights into the changes in plant morphology and stress responses. The advances in phenotyping software, analysis of plant growth dynamics, and identification of novel morphological responses will result in a deeper understanding of morphological reprogramming in response to salinity and other stresses. Studying plant growth reprogramming will bring us closer to understanding the link between changes in morphology and salt stress tolerance.

Acknowledgments

This work was supported by the Netherlands Organisation for Scientific Research, STW Learning from Nature project no. 10987. We thank Mark Tester for helpful suggestions on the manuscript.

References

- 1 FAO (2012) *FAO Statistical Yearbook*, FAO
- 2 Poljakoff-Mayber, A. and Gale, J. (1975) *Plants in Saline Environments*, Springer
- 3 Munns, R. and Tester, M. (2008) Mechanisms of salinity tolerance. *Annu. Rev. Plant Biol.* 59, 651–681
- 4 Katori, T. *et al.* (2010) Dissecting the genetic control of natural variation in salt tolerance of *Arabidopsis thaliana* accessions. *J. Exp. Bot.* 61, 1125–1138
- 5 Negrão, S. *et al.* (2013) New allelic variants found in key rice salt-tolerance genes: an association study. *Plant Biotechnol. J.* 11, 87–100
- 6 Demidchik, V. and Maathuis, F.J.M. (2007) Physiological roles of nonselective cation channels in plants: from salt stress to signalling and development. *New Phytol.* 175, 387–404
- 7 Shabala, S. and Cuin, T.A. (2008) Potassium transport and plant salt tolerance. *Physiol. Plantarum* 133, 651–669
- 8 Maathuis, F.J. and Sanders, D. (2001) Sodium uptake in *Arabidopsis* roots is regulated by cyclic nucleotides. *Plant Physiol.* 127, 1617–1625
- 9 Donaldson, L. *et al.* (2004) Salt and osmotic stress cause rapid increases in *Arabidopsis thaliana* cGMP levels. *FEBS Lett.* 569, 317–320
- 10 Yuan, F. *et al.* (2014) OSCA1 mediates osmotic-stress-evoked Ca²⁺ increases vital for osmosensing in *Arabidopsis* 514, 367–371
- 11 Peiter, E. *et al.* (2005) The vacuolar Ca²⁺-activated channel TPC1 regulates germination and stomatal movement. *Nature* 434, 404–408
- 12 Choi, W-G. *et al.* (2014) Salt stress-induced Ca²⁺ waves are associated with rapid, long-distance root-to-shoot signaling in plants. *Proc. Natl. Acad. Sci. U.S.A.* 111, 6497–6502
- 13 Shabala, S. *et al.* (2006) Extracellular Ca²⁺ ameliorates NaCl-induced K⁺ loss from *Arabidopsis* root and leaf cells by controlling plasma membrane K⁺-permeable channels. *Plant Physiol.* 141, 1653–1665
- 14 Sun, J. *et al.* (2010) H₂O₂ and cytosolic Ca²⁺ signals triggered by the PMH-coupled transport system mediate K⁺/Na⁺ homeostasis in NaCl-stressed *Populus euphratica* cells. *Plant Cell Environ.* 33, 943–958
- 15 Dubiella, U. *et al.* (2013) Calcium-dependent protein kinase/NADPH oxidase activation circuit is required for rapid defense signal propagation. *Proc. Natl. Acad. Sci. U.S.A.* 110, 8744–8749
- 16 Miller, G. *et al.* (2009) The plant NADPH oxidase RBOHD mediates rapid systemic signaling in response to diverse stimuli. *Sci. Signal.* 2, ra45
- 17 Jiang, C. *et al.* (2013) ROS-mediated vascular homeostatic control of root-to-shoot soil Na delivery in *Arabidopsis*. *EMBO J.* 32, 914
- 18 Jiang, C. *et al.* (2013) An *Arabidopsis* soil-salinity-tolerance mutation confers ethylene-mediated enhancement of sodium/potassium homeostasis. *Plant Cell* 25, 3535–3552
- 19 Sirichandra, C. *et al.* (2009) Phosphorylation of the *Arabidopsis* AtrbohF NADPH oxidase by OST1 protein kinase. *FEBS Lett.* 583, 2982–2986
- 20 Zhang, Y. *et al.* (2009) Phospholipase α 1 and phosphatidic acid regulate NADPH oxidase activity and production of reactive oxygen species in ABA-mediated stomatal closure in *Arabidopsis*. *Plant Cell* 21, 2357–2377
- 21 Ismail, A. *et al.* (2014) Life and death under salt stress: same players, different timing? *J. Exp. Bot.* 65, 2963–2979
- 22 Barrero, J.M. *et al.* (2006) Both abscisic acid (ABA)-dependent and ABA-independent pathways govern the induction of NCED3, AAO3 and ABA1 in response to salt stress. *Plant Cell Environ.* 29, 2000–2008
- 23 Ruiz-Sola, M.Á. *et al.* (2014) A root specific induction of carotenoid biosynthesis contributes to ABA production upon salt stress in *Arabidopsis*. *PLoS ONE* 9, e90765
- 24 Geng, Y. *et al.* (2013) A spatio-temporal understanding of growth regulation during the salt stress response in *Arabidopsis*. *Plant Cell* 25, 2132–2154
- 25 Umezawa, T. *et al.* (2013) Genetics and phosphoproteomics reveal a protein phosphorylation network in the abscisic acid signaling pathway in *Arabidopsis thaliana*. *Sci. Signal.* 6, rs8
- 26 Boudsocq, M. *et al.* (2004) Identification of nine sucrose nonfermenting 1-related protein kinases 2 activated by hyperosmotic and saline stresses in *Arabidopsis thaliana*. *J. Biol. Chem.* 279, 41758–41766
- 27 Testerink, C. *et al.* (2004) Isolation and identification of phosphatidic acid targets from plants. *Plant J.* 39, 527–536
- 28 McLoughlin, F. *et al.* (2012) The Snf1-related protein kinases SnRK2.4 and SnRK2.10 are involved in maintenance of root system architecture during salt stress. *Plant J.* 72, 436–449
- 29 Munnik, T. and Vermeer, J.E.M. (2010) Osmotic stress-induced phosphoinositide and inositol phosphate signalling in plants. *Plant Cell Environ.* 33, 655–669
- 30 Lemtiri-Chlieh, F. *et al.* (2003) Inositol hexakisphosphate mobilizes an endomembrane store of calcium in guard cells. *Proc. Natl. Acad. Sci. U.S.A.* 100, 10091–10095
- 31 McLoughlin, F. and Testerink, C. (2013) Phosphatidic acid, a versatile water-stress signal in roots. *Front. Plant Sci.* 4, 525
- 32 McLoughlin, F. *et al.* (2013) Identification of novel candidate phosphatidic acid-binding proteins involved in the salt-stress response of *Arabidopsis thaliana* roots. *Biochem. J.* 450, 573–581
- 33 Zhou, H. *et al.* (2014) Inhibition of the *Arabidopsis* salt overly sensitive pathway by 14-3-3 proteins. *Plant Cell* 26, 1166–1182
- 34 Halfter, U. *et al.* (2000) The *Arabidopsis* SOS2 protein kinase physically interacts with and is activated by the calcium-binding protein SOS3. *Proc. Natl. Acad. Sci. U.S.A.* 97, 3735–3740
- 35 Shi, H. *et al.* (2002) The putative plasma membrane Na⁺/H⁺ antiporter SOS1 controls long-distance Na⁺ transport in plants. *Proc. Natl. Acad. Sci. U.S.A.* 14, 465–477
- 36 Brady, S.M. *et al.* (2007) A high-resolution root spatiotemporal map reveals dominant expression patterns. *Science* 318, 801–806
- 37 Bassil, E. *et al.* (2011) The *Arabidopsis* intracellular Na⁺/H⁺ antiporters NHX5 and NHX6 are endosome associated and necessary for plant growth and development. *Plant Cell* 23, 224–239
- 38 Bassil, E. *et al.* (2011) The *Arabidopsis* Na⁺/H⁺ antiporters NHX1 and NHX2 control vacuolar pH and K⁺ homeostasis to regulate growth, flower development, and reproduction. *Plant Cell* 23, 3482–3497
- 39 Hernández, A. *et al.* (2009) Mutants of the *Arabidopsis thaliana* cation/H⁺ antiporter AtNHX1 conferring increased salt tolerance in yeast: the endosome/prevacuolar compartment is a target for salt toxicity. *J. Biol. Chem.* 284, 14276–14285
- 40 Roy, S.J. *et al.* (2014) Salt resistant crop plants. *Curr. Opin. Biotechnol.* 26, 115–124
- 41 Dubois, M. *et al.* (2013) Ethylene Response Factor6 acts as a central regulator of leaf growth under water-limiting conditions in *Arabidopsis*. *Plant Physiol.* 162, 319–332

- 42 Achard, P. *et al.* (2006) Integration of plant responses to environmentally activated phytohormonal signals. *Science* 311, 91–94
- 43 West, G. *et al.* (2004) Cell cycle modulation in the response of the primary root of *Arabidopsis* to salt stress. *Plant Physiol.* 135, 1050–1058
- 44 Chaumont, F. and Tyerman, S.D. (2014) Aquaporins: highly regulated channels controlling plant water relations. *Plant Physiol.* 164, 1600–1618
- 45 Boursiac, Y. *et al.* (2005) Early effects of salinity on water transport in *Arabidopsis* roots. Molecular and cellular features of aquaporin expression. *Plant Physiol.* 139, 790–805
- 46 Prak, S. *et al.* (2008) Multiple phosphorylations in the C-terminal tail of plant plasma membrane aquaporins. *Mol. Cell. Proteomics* 7, 1019–1030
- 47 Baxter, I. *et al.* (2009) Root suberin forms an extracellular barrier that affects water relations and mineral nutrition in *Arabidopsis*. *PLoS Genet.* 5, e1000492
- 48 Romero-Aranda, R. *et al.* (2001) Tomato plant-water uptake and plant-water relationships under saline growth conditions. *Plant Sci.* 160, 265–272
- 49 Tenhaken, R. (2014) Cell wall remodeling under abiotic stress. *Front. Plant Sci.* 5, 771
- 50 Weber, C. *et al.* (2008) Plant stress granules and mRNA processing bodies are distinct from heat stress granules. *Plant J.* 56, 517–530
- 51 Yan, C. *et al.* (2014) Tudor-SN, a component of stress granules, regulates growth under salt stress by modulating GA20ox3 mRNA levels in *Arabidopsis*. *J. Exp. Bot.* 65, 5933–5944
- 52 Moller, I.S. *et al.* (2009) Shoot Na⁺ exclusion and increased salinity tolerance engineered by cell type-specific alteration of Na⁺ transport in *Arabidopsis*. *Plant Cell* 21, 2163–2178
- 53 Essa, T.A. (2002) Effect of salinity stress on growth and nutrient composition of three soybean (*Glycine max* L. Merrill) cultivars. *J. Agron. Crop Sci.* 188, 86–93
- 54 Benito, B. *et al.* (2014) The twins K⁺ and Na⁺ in plants. *J. Plant Physiol.* 171, 723–731
- 55 Liu, J. and Zhu, J.K. (1997) An *Arabidopsis* mutant that requires increased calcium for potassium nutrition and salt tolerance. *Proc. Natl. Acad. Sci. U.S.A.* 94, 14960–14964
- 56 Chao, D.-Y. *et al.* (2013) Polyploids exhibit higher potassium uptake and salinity tolerance in *Arabidopsis*. *Science* 341, 658–659
- 57 Armengaud, P. *et al.* (2004) The potassium-dependent transcriptome of *Arabidopsis* reveals a prominent role of jasmonic acid in nutrient signaling. *Plant Physiol.* 136, 2556–2576
- 58 Chen, Q. *et al.* (2011) The basic helix-loop-helix transcription factor MYC2 directly represses PLETHORA expression during jasmonate-mediated modulation of the root stem cell niche in *Arabidopsis*. *Plant Cell* 23, 3335–3352
- 59 Kellermeier, F. *et al.* (2014) Analysis of the root system architecture of *Arabidopsis* provides a quantitative readout of crosstalk between nutritional signals. *Plant Cell* 26, 1480–1496
- 60 Julkowska, M.M. *et al.* (2014) Capturing *Arabidopsis* root architecture dynamics with ROOT-FIT reveals diversity in responses to salinity. *Plant Physiol.* 166, 1387–1402
- 61 Kellermeier, F. *et al.* (2013) Natural variation of *Arabidopsis* root architecture reveals complementing adaptive strategies to potassium starvation. *Plant Physiol.* 161, 1421–1432
- 62 Claeys, H. *et al.* (2014) What Is stress? Dose-response effects in commonly used in vitro stress assays. *Plant Physiol.* 165, 519–527
- 63 Pierik, R. and Testerink, C. (2014) The art of being flexible: how to escape from shade, salt and drought. *Plant Physiol.* 166, 5–22
- 64 Duan, L. *et al.* (2013) Endodermal ABA signaling promotes lateral root quiescence during salt stress in *Arabidopsis* seedlings. *Plant Cell* 25, 324–341
- 65 Petricka, J.J. *et al.* (2012) Control of *Arabidopsis* root development. *Annu. Rev. Plant Biol.* 63, 563–590
- 66 Moreno-Risueno, M.A. *et al.* (2010) Oscillating gene expression determines competence for periodic *Arabidopsis* root branching. *Science* 329, 1306–1311
- 67 Vanneste, S. *et al.* (2005) Cell cycle progression in the pericycle is not sufficient for SOLITARY ROOT/IAA14-mediated lateral root initiation in *Arabidopsis thaliana*. *Plant Cell* 17, 3035–3050
- 68 Parizot, B. *et al.* (2008) Diarch symmetry of the vascular bundle in *Arabidopsis* root encompasses the pericycle and is reflected in distich lateral root initiation. *Plant Physiol.* 146, 140–148
- 69 Ivanchenko, M.G. *et al.* (2006) Mutations in the Diageotropica (Dgt) gene uncouple patterned cell division during lateral root initiation from proliferative cell division in the pericycle. *Plant J.* 46, 436–447
- 70 Vermeer, J.E.M. *et al.* (2014) A spatial accommodation by neighboring cells is required for organ initiation in *Arabidopsis*. *Science* 343, 178–183
- 71 Li, X. *et al.* (2006) Cytokinin-mediated cell cycling arrest of pericycle founder cells in lateral root initiation of *Arabidopsis*. *Plant Cell Physiol.* 47, 1112–1123
- 72 Laplaze, L. *et al.* (2007) Cytokinins act directly on lateral root founder cells to inhibit root initiation. *Plant Cell* 19, 3889–3900
- 73 Zolla, G. *et al.* (2010) Mild salinity stimulates a stress-induced morphogenic response in *Arabidopsis thaliana* roots. *J. Exp. Bot.* 61, 211–224
- 74 Julkowska, M.M. *et al.* (2014) Identification and functional characterization of the *Arabidopsis* Snf1-related protein kinase SnRK2.4 phosphatidic acid-binding domain. *Plant Cell Environ.* 38, 614–624
- 75 Osmont, K.S. *et al.* (2007) Hidden branches: developments in root system architecture. *Annu. Rev. Plant Biol.* 58, 93–113
- 76 Galvan-Ampudia, C.S. *et al.* (2013) Halotropism is a response of plant roots to avoid a saline environment. *Curr. Biol.* 23, 2044–2050
- 77 Robins, N.E. *et al.* (2014) Beyond the barrier: communication in the root through the endodermis. *Plant Physiol.* 166, 551–559
- 78 Wang, Y. *et al.* (2008) Salt-induced plasticity of root hair development is caused by ion disequilibrium in *Arabidopsis thaliana*. *J. Plant Res.* 121, 87–96
- 79 Wu, S.J. *et al.* (1996) SOS1, a genetic locus essential for salt tolerance and potassium acquisition. *Plant Cell* 8, 617–627
- 80 Liu, J. *et al.* (2000) The *Arabidopsis thaliana* SOS2 gene encodes a protein kinase that is required for salt tolerance. *Proc. Natl. Acad. Sci. U.S.A.* 97, 3730–3734
- 81 Faiyue, B. *et al.* (2010) Studies on sodium bypass flow in lateral rootless mutants *lrt1* and *lrt2*, and crown rootless mutant *cr11of* rice (*Oryza sativa* L.). *Plant Cell Environ.* 33, 687–701
- 82 Faiyue, B. *et al.* (2010) The role of lateral roots in bypass flow in rice (*Oryza sativa* L.). *Plant Cell Environ.* 33, 702–716
- 83 Sunarpi, H.T. *et al.* (2005) Enhanced salt tolerance mediated by AtHKT1 transporter-induced Na⁺ unloading from xylem vessels to xylem parenchyma cells. *Plant J.* 44, 928–938
- 84 Maser, P. *et al.* (2002) Altered shoot/root Na⁺ distribution and bifurcating salt sensitivity in *Arabidopsis* by genetic disruption of the Na⁺ transporter AtHKT1. *FEBS Lett.* 531, 157–161
- 85 Achard, P. *et al.* (2008) Plant DELLAs restrain growth and promote survival of adversity by reducing the levels of reactive oxygen species. *Curr. Biol.* 18, 656–660
- 86 Drerup, M.M. *et al.* (2013) The Calcineurin B-like calcium sensors CBL1 and CBL9 together with their interacting protein kinase CIPK26 regulate the *Arabidopsis* NADPH oxidase RBOHF. *Mol. Plant* 6, 559–569